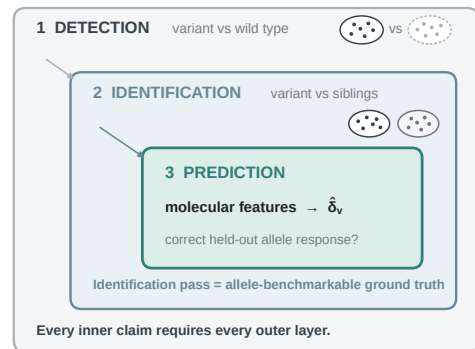


a

Allele-specific prediction is a nested solvability problem

A model can be judged only after the measurement detects a response and reproducibly identifies the allele.



Verdict: where is the binding limit?

Detection or identification fails

MEASUREMENT-LIMITED

Ground truth cannot reward an allele model.
Do not rank models at allele resolution.

Identification passes; model < ceiling

MODEL-LIMITED

A meaningful computational gap.

Model reaches the measurement ceiling

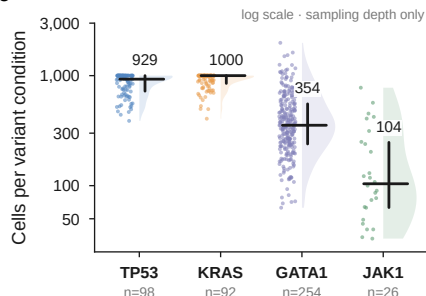
RESOLVED AT CURRENT CEILING

Prediction is supported at this resolution.

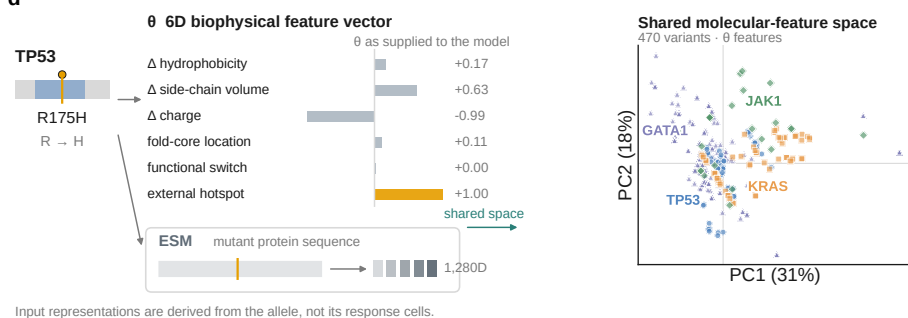
! Direction recovered (Pearson- δ) $\neq$ allele identified or predicted

Held-out task observed cells define δ_v and resolution • model sees only θ /ESM and predicts $\hat{\delta}_v$

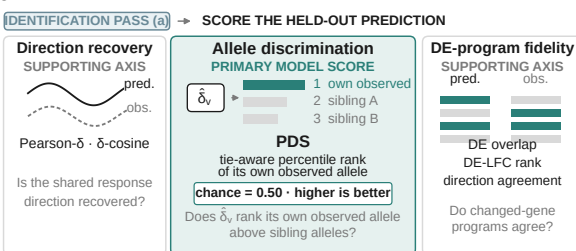
c



d

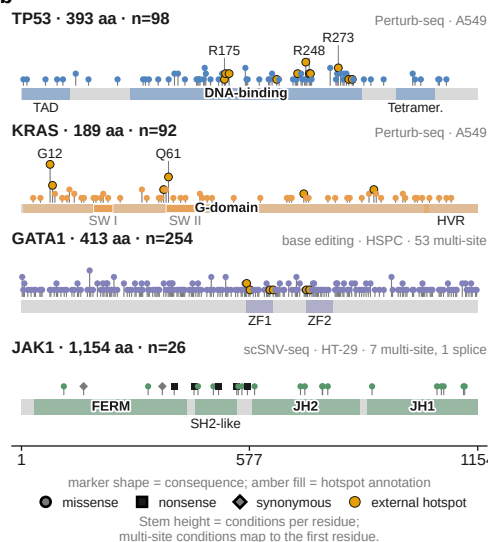


e



Measurement identification validates the benchmark; PDS evaluates the model within it.

b



f

